Human Capital Success Stories

How countries move from poor and short-lived to rich and long-lived

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General Assembly Data Science Bootcamp · Project 2: EDA

World Development Statistics, Gapminder · 1960 to 2020

Introduction

The foundation's problem

- A philanthropic foundation wants to design development programs that actually work.
- It has no clear blueprint, so it needs proven examples of countries that succeeded.

Our question

- Which countries truly moved from low income and short lives to high income and long lives between 1960 and 2020?
- How did they do it, and what can the foundation copy?

The Data

Source and scope

- Four datasets from Gapminder: population, life expectancy, income (GNI per capita), and fertility.
- Every country from 1960 to 2020, one value per country per year.

Why these four, and why kept separate

- Income and life expectancy show the standard of living. Population and fertility show how families and societies change.
- Fertility is an early signal: it starts falling before income rises, so it flags countries entering their transition, and family planning is a lever the foundation can fund.
- We could merge the four into one score (for example with PCA), but we kept them separate so the foundation can see each lever and target the right program.

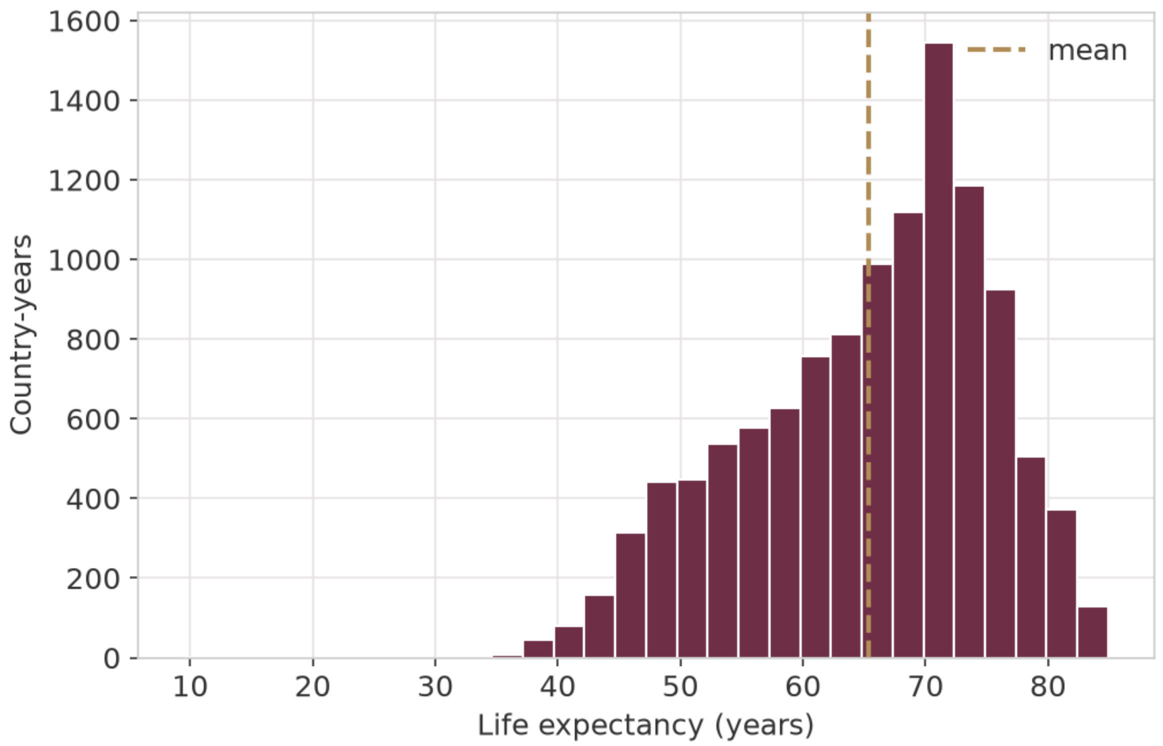
Cleaning the Data

Preparing the data

- Reshaped each file from wide to long, and turned text like 3.28M into real numbers.
- Merged the four files into one clean table of 190 countries and 11,584 rows.
- A few small states dropped in the merge, but 190 is plenty to see the trends. It would only matter if we asked about those specific countries.

A data quality check

- Life expectancy is left skewed. The mean (65.3) sits below the median (67.3).
- The lowest values, like Rwanda in 1994 (near 9.5 years), come from real events like genocide and starvation, not data errors.

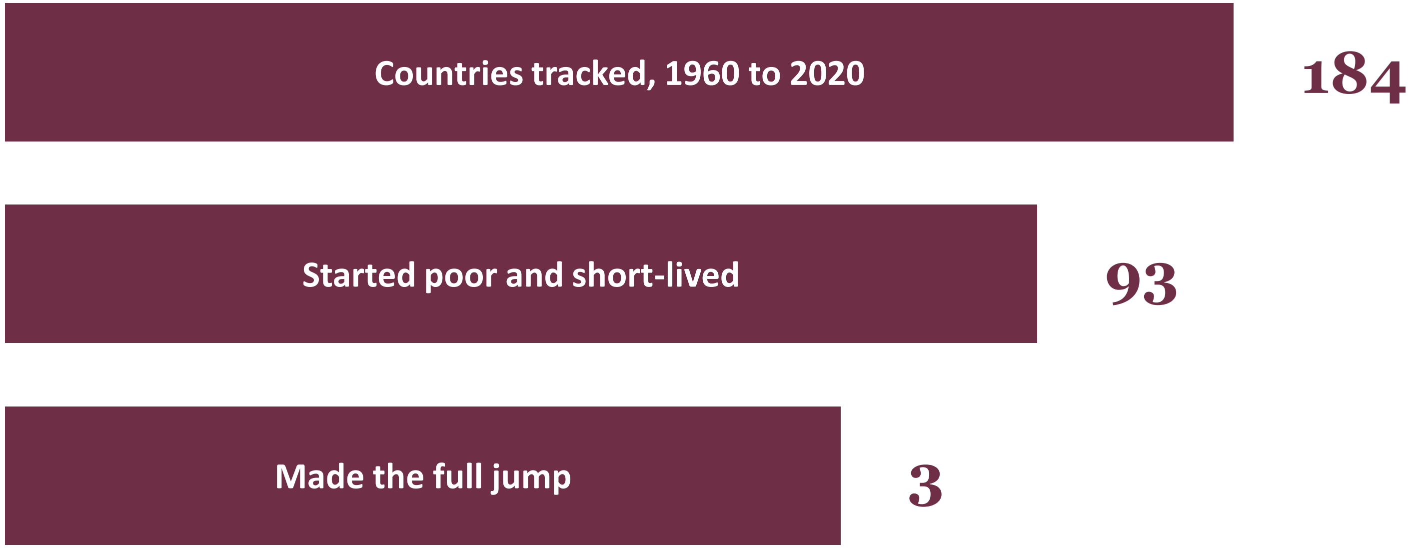


We keep these crisis years because they are realistic. Removing them would hide real history and bias the trend.

How We Found the Transition Countries

The rule we used

- A full transition means starting poor and short-lived in 1960 (income under 2,000 and life under 60),
- and ending rich and long-lived by 2020 (income over 10,000 and life over 72).

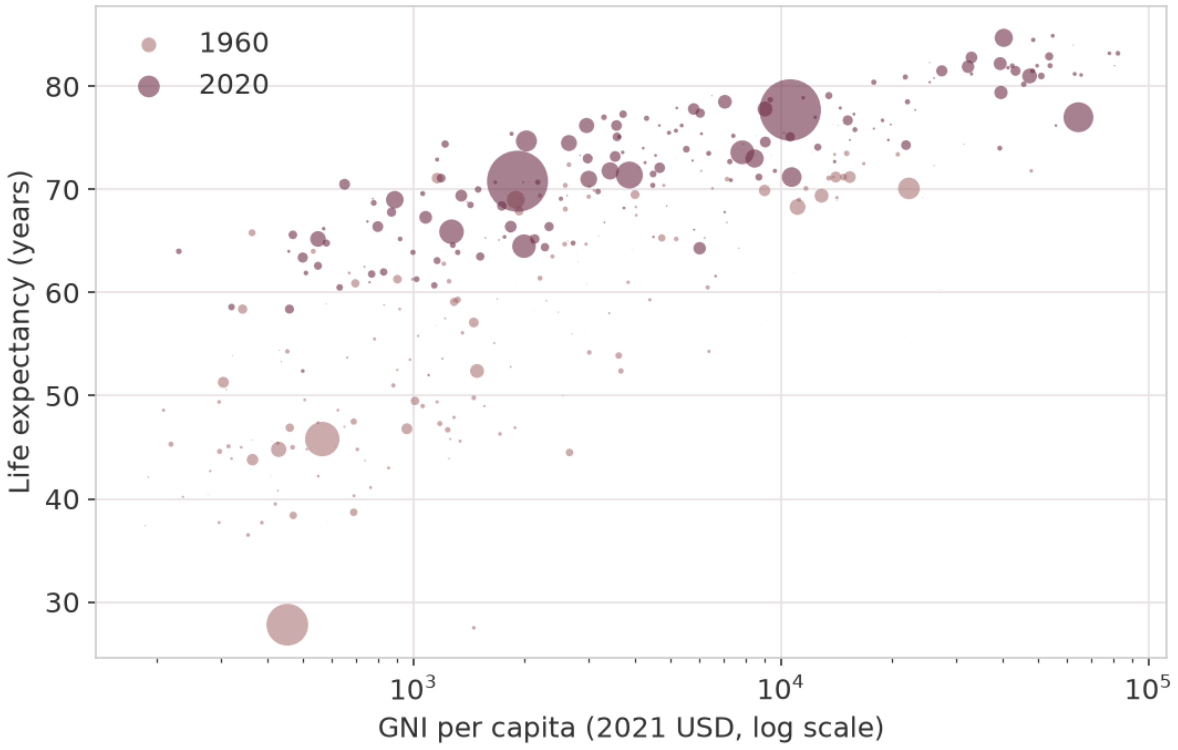


The full leap is rare, only three of 184 made it. Rare success means the foundation should copy proven models, not gamble on luck.

Income and Life Expectancy

Higher income comes with longer life

- Income is on a log scale, so the poorer countries spread out instead of bunching on the left.
- From 1960 (rose) to 2020 (maroon) the whole world shifted up and to the right.
- Bubble size is population.

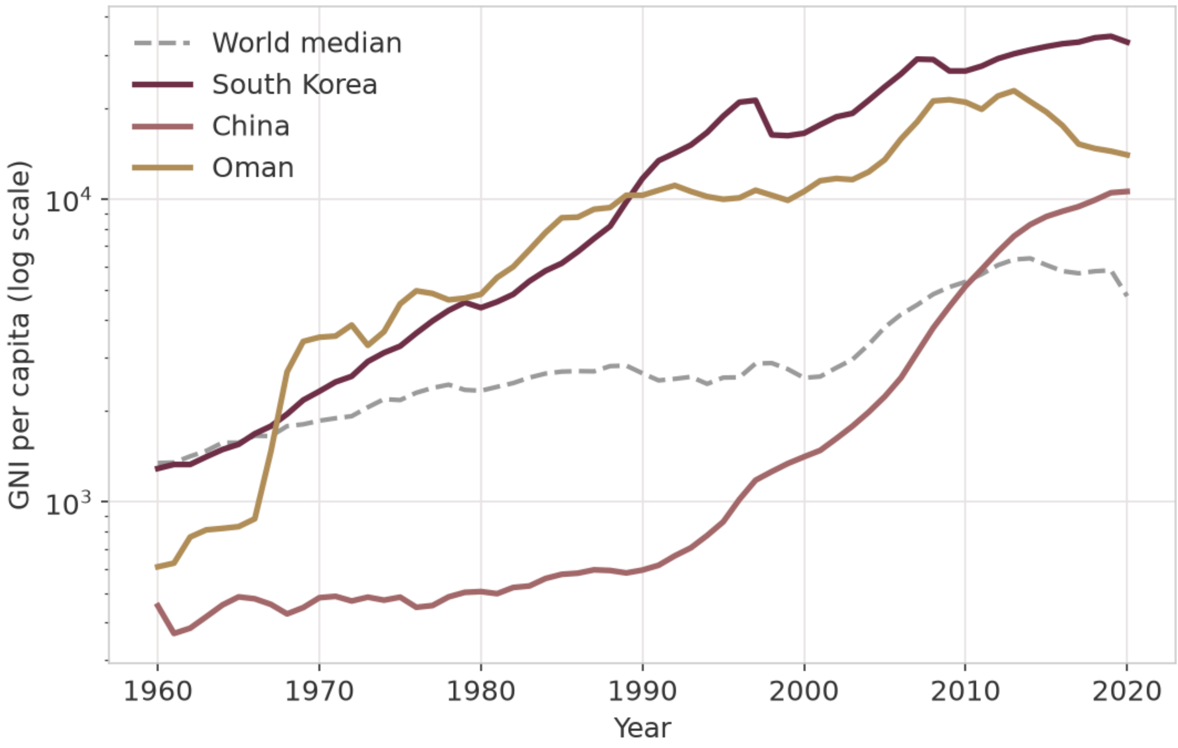


Quality of life is tied to income (r = 0.82). So to add years of life, the foundation must also raise income, through industry and education.

The Three Transition Countries

Three countries, three different decades

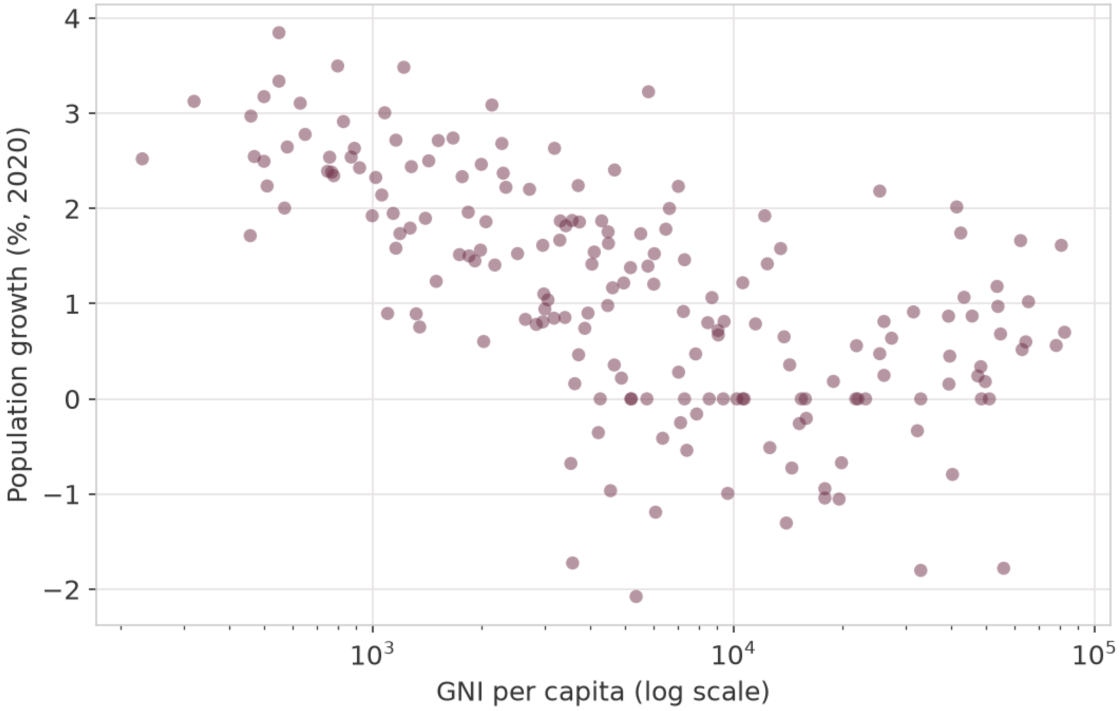
- Their income grew 22.9 to 25.6 times, far above the world median (1,345 to 4,785).
- Oman took off on oil around 1970, a sudden jump but volatile after 2010.
- Korea rose on industry and education, China after its reforms in the 2000s.



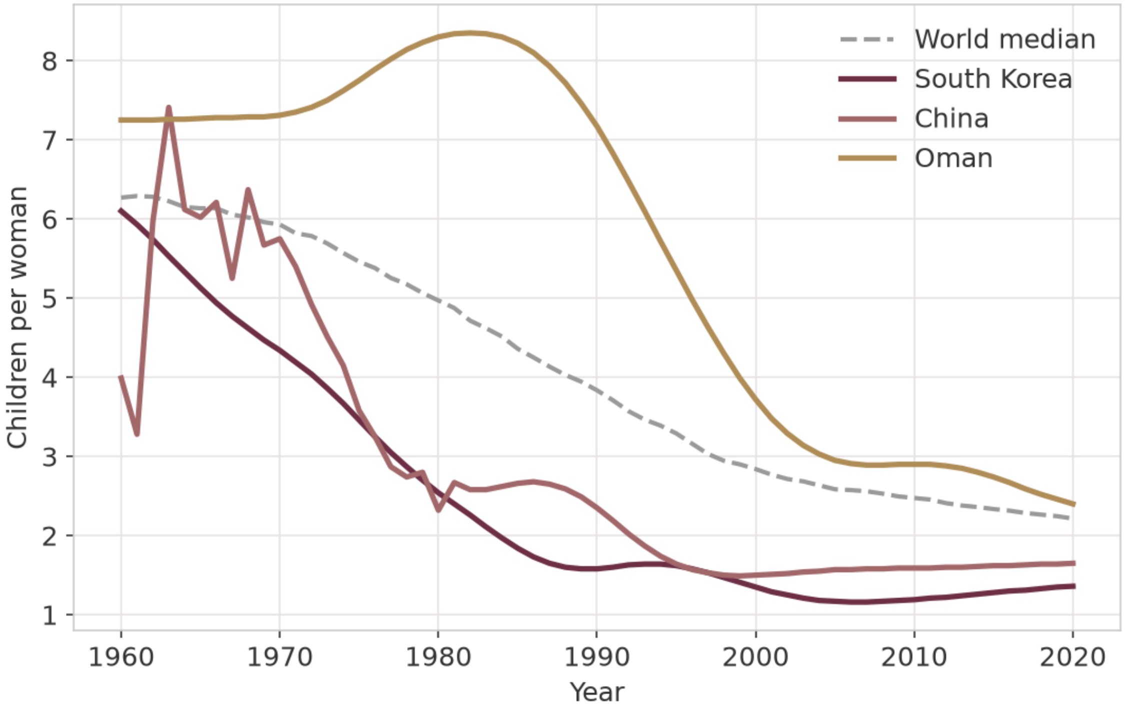
Each took off with a different trigger: oil, industry, or reform. The foundation should fit its program to the country, not use one plan for all.

Population and Fertility

Population grows more slowly



Families have fewer children

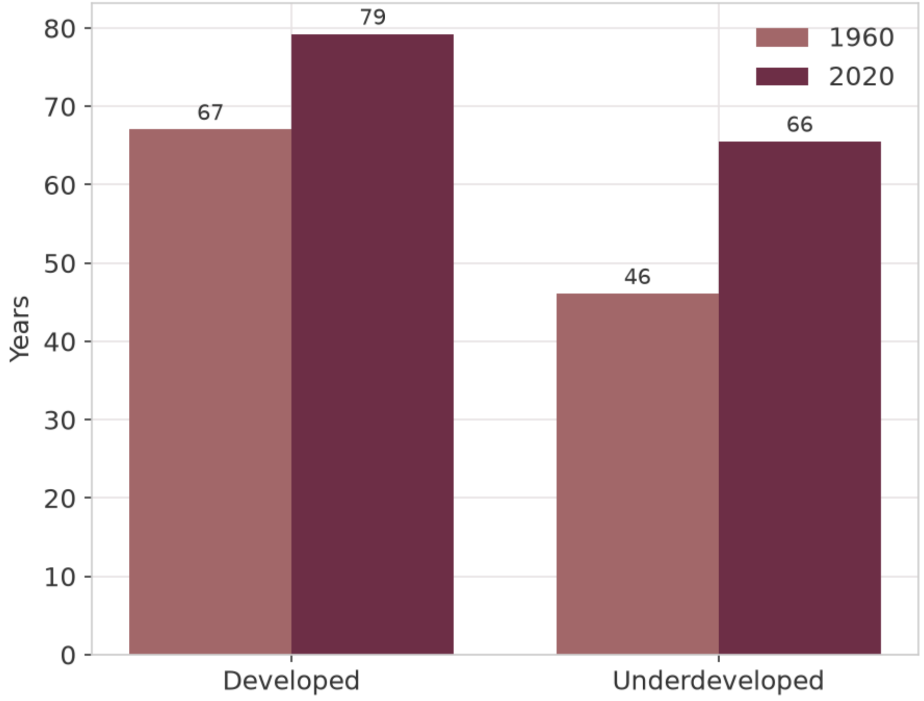


Fewer children signal real development, fertility fell from about 6 to under 2. Family planning and girls' schooling are levers the foundation can fund to speed it.

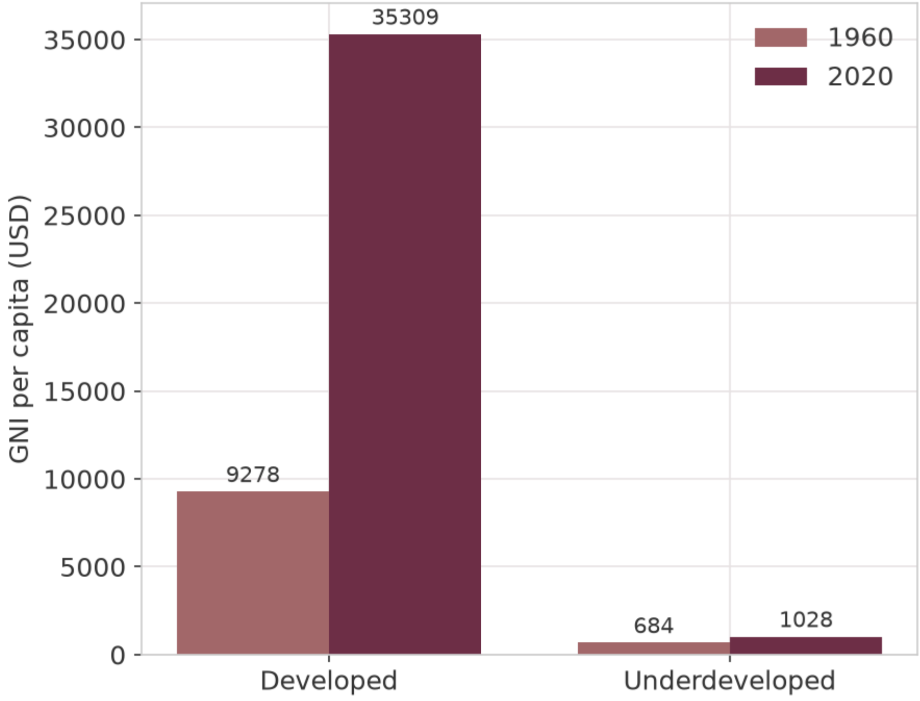
Developed vs Underdeveloped

Group averages (mean). Developed = income over 12,000 in 2020, underdeveloped = under 2,000.

Life expectancy converged



But income diverged



The gains were not shared equally. Health is cheap, so fund it everywhere first. Wealth needs industry and skills, so invest where a country is ready.

Why It Happened

One development process ties every finding together

- Incomes rose through export-led growth and heavy investment in education, the East Asian Miracle [1].
- Longer lives came from public health: clean water, sanitation, vaccines and antibiotics [2].
- Families had fewer children as girls got more schooling and child mortality fell [3, 4, 5].
- Together these three form the demographic transition, the shift every developed country went through.
- These same three drivers, education, public health and industry, are exactly the programs the foundation should fund.

Key Insights

- The full transition is rare. Only 3 of 190 countries made it: South Korea, China and Oman.
- Health reached almost everyone, but income only pulled ahead where countries built industry and skills.
- As countries develop, families shrink. Fertility fell from about 6 to under 2 children per woman.
- Income and life expectancy are still tightly linked in 2020, with a correlation of 0.82.

Recommendations for the Foundation

- Copy the Korea model: fund export industry paired with mass education. Why: it is the most durable path, and Korea kept rising for 60 years.
- Fund low-cost health first: vaccination, clean water, and clinics. Why: these add years of life fast and reach poor countries too.
- Invest in girls' schooling and family-planning clinics. Why: educated women have smaller families, the engine of the whole transition.
- For oil economies like Oman, fund diversification into industry and skills. Why: oil income is volatile and Oman's fell after 2010.
- Target the poorest countries first. Why: each dollar buys the most years of life where health is still low.

Summary

- Since 1960 the world's health converged while its wealth diverged.
- Only three countries made the full leap from poor and short-lived to rich and long-lived, each triggered by a different policy.
- Development came with slower population growth and far fewer children per family.
- For the foundation, the lesson is clear: fund health and education everywhere, and copy the proven paths to income.

Development is possible in any decade. The foundation's job is to trigger it.

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Thank you